

2 Experimental works

All the reagents used in this study were utilized without further purification or unless otherwise specified.

2.1 Reagents

- Monolith Silica (specified)
- Distilled Water
- Para-aminobenzoic acid (PABA). (specified)
- Anhydrous Methanol
- Anhydrous Ethanol
- 20 layers Pre-synthesized HKUST-1 over ZP (specified)

2.2 Preparation of Adsorbent

Before use, the crucible was dried overnight in an oven to ensure a clean and dry surface. A stoichiometric amount of monolithic silicas was placed in the crucible and subjected to calcination for straight 08 hours at 480 Celsius temperatures in the wise-Therm muffle furnace for the improvement and enhancement of the silica for further silica experimentation and procedure.

In the next step, a known stoichiometric amount of p-aminobenzoic acid was introduced into the 250ml round bottom flask filled with distilled water and stirred it for 5 minutes to get a homogenous solution after complete dissolution. The previously calcinated silica was then added to this solution under continuous stirring to get a homogenous and dispersed solution.

This dispersed solution was refluxed for straight 30 mint under controlled thermal condition to have a strong interaction between monolith silica and PABA. The reflux proces was done using a laboratory assembled reflux setup comprising a heating mantle, a cooling condenser connected by soft tubing and continuous water supply and oil bath.

The round bottom flask was dipped in the oil bath containing the solution and fitted with the condenser. The continuous thermal heat provided by the oil bath initiate the boiling while the condenser condensed the vapour and return it back to the mixture and hence maintain the reflux process throughout the experiment.

After the reflux, the rection setup was allowed to cool down naturally for 20-30 min. The solution mixture was transferred to the beaker and covered with aluminum foil and left

undisturbed for 24h at room temperature. Following the 24h standing period, the formation of crystal was observed. To dissolve these formed crystals, 20 ml of methanol was added gradually under stirring conditions. This process was repeated three times until the crystals were fully dissolved. The excess methanol was removed carefully with the help of a syringe. The residual mother liquid or saturated jelly was dried in oven until it became powder. A silica-amine functionalized material was obtained as a homogeneous fine powder.

Approximately 0.25g of functionalized silica was dispersed in 5 ml ethanol followed by ultrasonication for 5 minutes at ambient temperature for the activation of functionalized silica-amine powder. The pre-synthesized stoichiometric amount of HKUST-1@ZP (20 layered HKUST-1 over ZP nano-platform) was introduced into ultrasonically dispersed functionalized silica-amine suspension and further sonicated for the straight 30 min at ambient temperature to obtain desired MOF composite (HKUST-1@ZP/silica).

2.3 Sorption of Carbon dioxide Gas

To study the gas uptake or sorption of the MOF, gases are passed through the MOF by a lab assembled set-up consisting of a u-shaped glass sample holder sealed with a gas tighter stopper, a needle valve to control the gas flow and pressure an external heating system and a gas cylinder for the continuous flow of gases i.e CO₂ N₂ gas.

The MOF gas sorption performance was checked and evaluated by studying it on different cycles of temperature and pressure.

Before each cycle, the MOF was desorbed for maximum carbon dioxide uptake. The glass u-shaped sample is filled with MOF and the initial weight of this whole set-up was measured. The carbon dioxide gas was passed through it at ambient temperature for 30 minutes. The set-up was again weighted, and the difference in the initial and final weight was noted which determined the amount of carbon dioxide uptake by the MOF gravimetrically in mmol.

This process was repeated at ambient temperature consecutively until the gas uptake mass became constant (mass equilibrium is achieved). This was the maximum gas uptake by the MOF. The MOF was desorbed and weighed for next cycle.

To investigate the influence of temperature on